Lecture 13: Linear Regression

IEOR E4709: Statistic Analysis and Time Series
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1 Annoucement

Midterm exam will be on Wednesday, March 12, 10:00-11:30am. Sample exam and midterm review sheet have been posted on Courseworks in a folder named "Midterm". Solutions to the sample exam will be posted on March 6 by 5pm. There will be a midterm review session conducted by the TA, Boxuan Li on Friday, March 7, 12-1:30pm, in 523 Mudd.

2 How to choose priors in Bayesian analysis?

One possibility is to choose a prior which conjugates to the likelihood. Another possibility is to choose subjective priors. A third possibility is to estimate priors from data, following an empirical Bayesian approach.

If you have no knowledge of the parameter a priori, a fourth possibility is to choose a non-informative prior. For instance, assume $X \sim Bin(n, p)$. A plausible noninformative prior on the parameter p would be $p \in U[0, 1]$, i.e., uniform on the $[0, 1]$ interval. However, by doing so, we would obtain that $\sqrt{p}$ has a higher density near one than near 0. This is because the square root of a number in $(0, 1)$ is larger than the number itself, thus we know intuitively that the distribution of $\sqrt{p}$ must concentrate its probability toward the right end of the interval $(0, 1)$. (Specifically, the density function would be $f_{\sqrt{p}}(x) = 2x$ for $0 < x < 1$). Thus *ignorance* about p seems to lead to *knowledge* about $\sqrt{p}$, which is paradoxical. We would like the prior to be invariant under reparametrization. We will achieve this by using the Fisher information.

Recall that the Fisher information equals $I(\theta) = \mathbb{E}_\theta \left[\left(\frac{\partial \log f(X|\theta)}{\partial \theta} \right)^2 \right]$. Let us define the *Jeffreys prior* as $\pi(\theta) \propto I(\theta)^{1/2}$. Consider a model $X \sim f(x|\theta)$, and its reparametrized version $X \sim g(x|\eta)$, where $\eta = h(\theta)$ and h is a differentiable monotone transformation of θ. Applying the chain rule,

we would have that

$$\begin{aligned} I(\theta) &= \int \left(\frac{\partial}{\partial \theta} \log f(x|\theta) \right)^2 f(x|\theta) dx \\ &= \int \left(\frac{\partial}{\partial \theta} \log g(x|\eta) \right)^2 g(x|\eta) dx \\ &= \int \left(\frac{\partial}{\partial \eta} \log g(x|\eta) \Big|_{\eta=h(\theta)} h'(\theta) \right)^2 g(x|h(\theta)) dx \\ &= (h'(\theta))^2 I(h(\theta)). \end{aligned}$$

Next, recall that if X is a r.v. with pdf $f_X(x)$ that is continuous on its support, and $Y = g(X)$ where g is a monotone function of X, then the pdf f_Y of Y is given by

$$f_Y(y) = f_X(g^{-1}(y)) | \frac{\partial}{\partial y} g^{-1}(y) |$$

In our case $y = h(\theta)$, hence

$$\pi_Y(y) = \pi_x(h^{-1}(y)) \big| \frac{d}{dy} h^{-1}(y) \big|,$$

thus

$$\pi(h(\theta)) = \pi(\theta) \big| \frac{d}{dh(\theta)} h^{-1}(h(\theta)) \big| = \pi(\theta) \left| \frac{d\theta}{dh(\theta)} \right| = \frac{\pi(\theta)}{|h'(\theta)|}.$$

Hence, for the Jeffreys prior $\pi(\theta)$, we have

$$\pi(h(\theta)) = \pi(\theta) \left| h'(\theta) \right|^{-1} \propto I(\theta)^{1/2} \left| h'(\theta) \right|^{-1} = I(h(\theta))^{1/2}$$

and thus the prior is invariant under reparametrization. The Jeffreys prior favors values of θ for which $I(\theta)$ is large. Hence, it minimizes the effect of the prior distribution relative to the information in the data.

Example: Let $X \sim Bin(n, p)$, where n is known, so that $f(x|p) = \binom{n}{x} p^x (1-p)^{n-x}$. Then

$$\frac{\partial^2 \log f(x|p)}{\partial^2 p} = -\frac{x}{p^2} - \frac{n-x}{(1-p)^2}.$$

Taking expectation and multiplying by -1 (we are using the equivalent representation $I(p) = -\mathbb{E}_p \left[\frac{\partial^2 \log f(X|p)}{\partial^2 p} \right]$, we obtain: $I(p) = \frac{n}{p} + \frac{n}{1-p} = \frac{n}{p(1-p)}$. Thus the Jeffreys prior is

$$\pi(p) \propto (p(1-p))^{-1/2},$$

which we recognize as the $Beta_{(1/2,1/2)}$ distribution.

3 Simple Linear Regression

Consider a linear model

$$y = \alpha + \beta x,$$

where x is the *explanatory variable*, and y is the *response variable/dependent variable*. Suppose we are confronted with the following question: given observed data $(x_1, y_1), \ldots, (x_n, y_n)$, estimate α and β. The most direct approach is to minimize the ordinary least square (OLS), i.e.,

$$\min_{\alpha,\beta} \sum_{i=1}^{n} (y_i - (\alpha + \beta x_i))^2.$$

By setting the first order derivative to zero, it is straightforward to see that we obtain

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}, \qquad \hat{\beta} = S_{xx}^{-1} S_{xy} = Cov(x, y)/Var(x), \tag{1}$$

where $\bar{y} := \frac{1}{n}\sum_{i=1}^{n} y_i$, $\bar{x} := \frac{1}{n}\sum_{i=1}^{n} x_i$ are the sample means of the response variables, and of the explanatory variables, respectively. The quantities $S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2$, and $S_{xy} = \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$. We also use $Cov(x, y)$ to denote the sample covariance of x and y, and $Var(x)$ to denote the sample variance of x.

Next, consider the statistical model

$$y_i = \alpha + \beta x_i + \epsilon_i,$$

where ϵ_i is a zero-mean random variable. This model is called a *simple linear regression model*. It is a linear model because $\mathbb{E}[y_i|x_i] = \alpha + \beta x_i$, that is the *best linear* estimate for y_i given x_i, $\mathbb{E}[y_i|x_i]$, is linear in x_i.

Let us make the following assumptions: ϵ_i's have equal variance, ϵ_i's are independent, ϵ_i's have zero mean and variance σ^2. We define $\hat{y}_i := \hat{\alpha} + \hat{\beta} x_i$ as the *prediction* of the response given the explanatory variable x_i. The quantity $\hat{\epsilon}_i = y_i - \hat{y}_i$ is called the *residual*.

In linear regression models, the explanatory variable x is assumed to be observed (known) and not random. The linear model describes the relation between the mean of the response variable and the controlled explanatory variable. Consider the OLS estimates $\hat{\alpha}$ and $\hat{\beta}$ in Eq. (1). We can show that both of them are unbiased estimates of α and β in the simple linear regression model. Indeed

$$\begin{aligned}
\mathbb{E}\left[\hat{\beta}\right] &= \mathbb{E}\left[S_{xx}^{-1} S_{xy}\right] = S_{xx}^{-1}\mathbb{E}\left[S_{xy}\right] \\
&= S_{xx}^{-1} \sum_{i=1}^{n} (x_i - \bar{x})(\mathbb{E}[y_i] - \mathbb{E}\left[\bar{y}\right]) \\
&= S_{xx}^{-1} \sum_{i=1}^{n} (x_i - \bar{x})\left((\alpha + \beta x_i) - (\alpha + \beta\bar{x})\right) \\
&= S_{xx}^{-1} \beta S_{xx} \\
&= \beta.
\end{aligned}$$

Moreover,

$$\mathbb{E}\left[\hat{\alpha}\right] = \mathbb{E}\left[\bar{y}\right] - \mathbb{E}\left[\hat{\beta}\right]\bar{x} = \alpha + \beta\bar{x} - \beta\bar{x} = \alpha.$$

Assume further that ϵ_i's are Gaussian and independent. It can be shown that the estimators $\hat{\alpha}$ and $\hat{\beta}$ are jointly normal with variances and covariances given by

$$\begin{aligned}
Var(\hat{\beta}) &= S_{xx}^{-1}\sigma^2 \\
Cov(\hat{\alpha}, \hat{\beta}) &= -\bar{x} S_{xx}^{-1}\sigma^2 \\
Var(\hat{\alpha}) &= \frac{\sigma^2}{n} + \bar{x}^2 S_{xx}^{-1}\sigma^2.
\end{aligned} \tag{2}$$

Notice that the variances of $\hat{\alpha}$ and $\hat{\beta}$ and their covariance all depend on the unknown parameter σ^2. As σ is usually unknown, we would need to estimate σ^2 from the data. Define the sum of the squares of residuals

$$S_{rsdl} := \sum_{i=1}^{n} \hat{\epsilon}_i^2$$

It holds that that $\frac{S_{rsdl}}{\sigma^2}$ follows a χ^2 distribution with $n-2$ degrees of freedom. The number of degrees of freedom is equal to the number of observations minus the number of estimated regression coefficients. Notice that the expected value of a χ^2 distribution with k degrees of freedom is k. Hence, the estimator $\hat{\sigma}^2_{rsld} := \frac{S_{rsdl}}{n-2}$ is such that $\mathbb{E}\left[\hat{\sigma}^2_{rsld}\right] = \sigma^2$, i.e. it is an unbiased estimator of σ^2. Moreover, it can be proven that $\hat{\alpha}$ and $\hat{\beta}$ are independent of S_{rsdl} (next homework problem).

Replacing σ^2 with $\hat{\sigma}^2_{rsld}$ in the expression for the variances of Eq. (2), we obtain unbiased estimates of the variances of $\hat{\alpha}$ and $\hat{\beta}$ (This can be easily checked). Such an estimate of the standard deviation is called the *standard error* (s.e.) of the estimate. Thus, the standard errors of $\hat{\alpha}$ and $\hat{\beta}$ are

$$\begin{aligned} s.e.(\hat{\alpha}) &= \hat{\sigma}_{rsld}\sqrt{\left(\frac{1}{n} + \bar{x}^2 S_{xx}^{-1}\right)} \\ s.e.(\hat{\beta}) &= \hat{\sigma}_{rsld}\sqrt{S_{xx}^{-1}}, \end{aligned}$$

where $\hat{\sigma}_{rsld} = \sqrt{\hat{\sigma}^2_{rsld}}$. The standard errors of $\hat{\alpha}$ and $\hat{\beta}$ are a measure of how precisely the intercept and the slope have been estimated.

4 Hypothesis Testing and Confidence Intervals for Simple Linear Regression

Under the normality assumption of ϵ_i's, the statistic $t_1 := \frac{\hat{\beta}-\beta}{s.e(\hat{\beta})}$ follows a Student's t distribution with $n-2$ degrees of freedom (there are $n-2$ degrees of freedom because we are estimating two regression coefficients). Similarly, we have that $t_2 := \frac{\hat{\alpha}-\alpha}{s.e(\hat{\alpha})}$ follows a student's t distribution with $n-2$ degrees of freedom. This allows us to construct confidence intervals and do hypothesis testing.

Assume we want to test the null hypothesis $\beta = 0$ versus the alternative hypothesis $\beta \neq 0$. Under the null hypothesis, the test statistics $t_1 = \frac{\hat{\beta}}{s.e(\hat{\beta})}$ is distributed as a Student's t random variable with $n-2$ degrees of freedom. The test is carried out by comparing the observed value of the statistics t_1 with the critical value obtained from the t-table, which is $t_{n-2,\frac{\varrho}{2}}$, where ϱ is the specified significance level. (We are using ϱ as opposed to α which was used in previous lectures, because here α is being used to denote the intercept in the regression model). Here, we recall that $t_{n-2,\frac{\varrho}{2}}$ is the $\left(1-\frac{\varrho}{2}\right)$ percentile of a t distribution with $n-2$ degrees of freedom. Notice also that we divided by 2 because we are considering a two-sided alternative hypothesis. Consequently, the null hypothesis is rejected at the significance level ϱ if

$$|t_1| \geq t_{n-2,\frac{\varrho}{2}}$$

Similar criteria can be used to construct confidence intervals for the regression coefficients. More precisely, the $(1-\varrho)\times 100\%$ confidence interval for α is given by

$$\hat{\alpha} \pm t_{n-2,\frac{\varrho}{2}} \times s.e.(\hat{\alpha})$$

Similarly, the $(1-\varrho)\times 100\%$ confidence interval for β is given by

$$\hat{\beta} \pm t_{n-2,\frac{\varrho}{2}} \times s.e.(\hat{\beta})$$